Topical Antibiotic Use and Antimicrobial Resistance in *Cutibacterium acnes*

A Literature Review and Descriptive Data Analysis

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Abstract

Antibiotic resistance in *Cutibacterium acnes* is an important concern in acne treatment, particularly with the use of topical antibiotics such as clindamycin. This project examined antibiotic resistance patterns, genetic mutations associated with resistance, and the effects of combination treatments using published data from four studies. Data were organized and analyzed using Python, Pandas, and Matplotlib to calculate descriptive percentages and visualize treatment outcomes. The findings showed that resistance varied across antibiotics, with higher resistance percentages for erythromycin and clindamycin than for doxycycline and minocycline. Mutations in 23S rRNA were also associated with antibiotic resistance. In treatment comparisons, clindamycin combined with benzoyl peroxide showed lower resistant bacterial counts and greater acne improvement than clindamycin alone. However, resistant bacteria remained detectable in some participants after 24 weeks of combination treatment. Overall, the findings suggest that combination treatments may offer benefits over antibiotic monotherapy, but further research is needed to understand their long-term effects on antibiotic resistance.

1. Introduction

Acne vulgaris is a common skin condition that affects many adolescents and adults. *Cutibacterium acnes* (*C. acnes*) is a bacterium naturally found on the skin that plays a role in acne development. Topical antibiotics, such as clindamycin, are used to treat inflammatory acne. However, antibiotic resistance is a concern because resistant bacteria may respond less effectively to treatment (Zhu et al., 2025).

Previous studies have examined antibiotic resistance in *C. acnes* from different perspectives. Zhu et al. (2025) reviewed resistance patterns across published studies, while Ross et al. (1997) investigated genetic mutations associated with resistance to certain antibiotics. Together, these studies provide information about the frequency of resistance and its possible genetic causes.

Researchers have also studied combination treatments that include antibiotics and other acne medications. Cunliffe et al. (2002) compared clindamycin alone with clindamycin combined with benzoyl peroxide. More recently, Ghannoum et al. (2025) examined changes in *C. acnes* detection and clindamycin resistance during treatment with clindamycin, adapalene, and benzoyl peroxide.

This independent research project uses published data and Python to examine antibiotic resistance patterns, resistance-related genetic mutations, and outcomes of topical acne treatments. By analyzing findings from four studies, the project aims to better understand how antibiotic resistance relates to acne treatment and what the available evidence suggests about combination therapies.

2. Methods

2.1 Study Design and Data Sources

This project combined a focused literature review with descriptive analysis of published data. Four studies were selected to examine different aspects of antibiotic resistance in *Cutibacterium acnes*: topical treatment outcomes, resistance rates across studies, genetic mutations, and changes in bacterial resistance during combination treatment (Cunliffe et al., 2002; Ghannoum et al., 2025; Ross et al., 1997; Zhu et al., 2025). All data came from published tables, figures, or supplementary materials. No new patient data was collected.

2.2 Data Extraction and Organization

Relevant data were organized into tables using Python and Pandas. For Cunliffe et al. (2002), I extracted changes in resistant bacterial counts and acne lesion counts over 16 weeks. For Zhu et al. (2025), I organized antibiotic resistance data from 23 studies involving 2,046 *C. acnes* isolates.

For Ross et al. (1997), I recorded 23S rRNA mutations and antibiotic susceptibility measurements. For Ghannoum et al. (2025), I organized bacterial detection, clindamycin susceptibility, and acne lesion data from 45 participants who completed the study.

2.3 Data Analysis

I used Python, Pandas, and Matplotlib in Google Colab to calculate descriptive percentages, compare reported values, and create graphs.

For the treatment studies, I compared changes in bacterial counts, resistance status, and acne lesions across the reported time points. For the resistance review, I calculated aggregate resistance percentages by dividing the total number of resistant isolates by the total number of isolates in studies reporting each antibiotic. These calculations were separate from the pooled estimates in the original meta-analysis (Zhu et al., 2025).

For the genetic analysis, I counted the reported mutations and examined their relationship with antibiotic susceptibility (Ross et al., 1997).

All analyses were descriptive. I did not conduct new clinical experiments, perform new statistical significance tests, or calculate a new meta-analysis.

3. Results

3.1 Clindamycin Monotherapy vs. Combination Treatment

The analysis of Cunliffe et al. (2002) showed differences between clindamycin alone and clindamycin combined with benzoyl peroxide (BPO). At Week 16, the mean log10 change in clindamycin-resistant *C. acnes* counts was +1.2 with clindamycin alone and −0.2 with combination treatment. The original study reported a statistically significant difference between the groups ($p = 0.018$) (Cunliffe et al., 2002).

Combination treatment also showed greater improvement in acne lesions. At Week 16, the median reduction in total lesions was 52.7% with combination treatment, compared with 27.5% with clindamycin alone (Cunliffe et al., 2002).

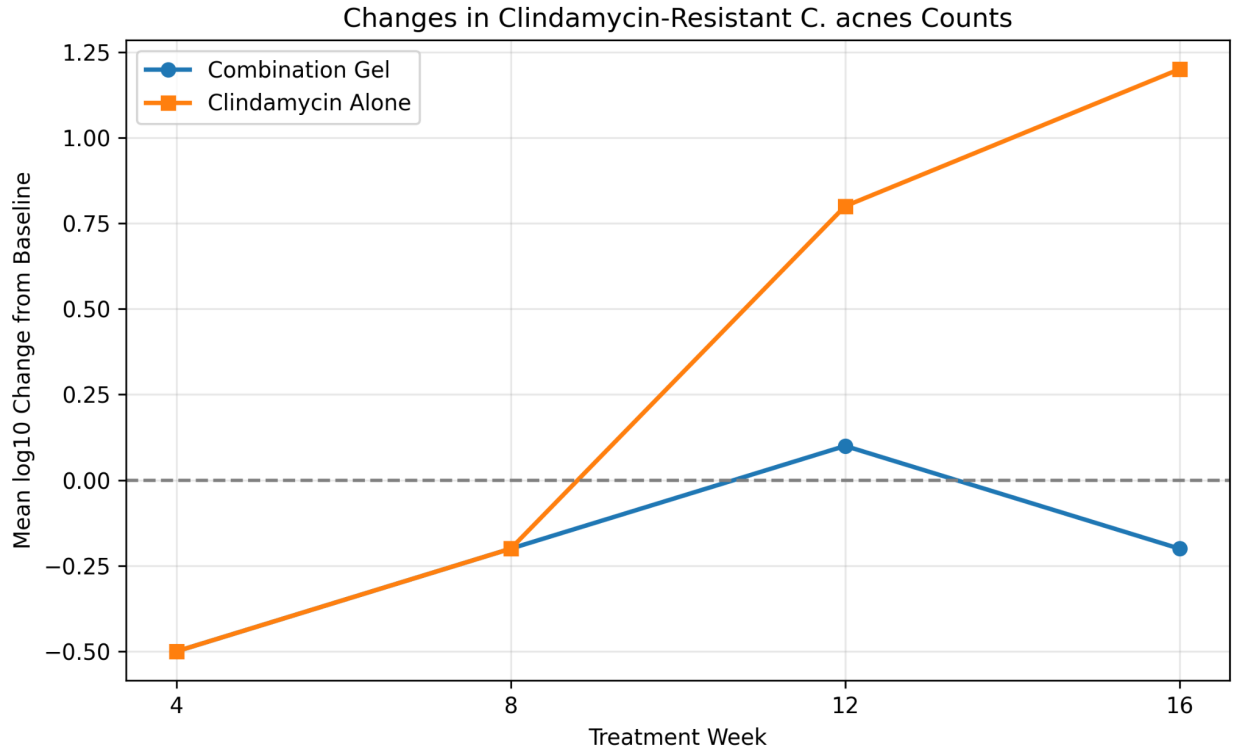


Figure 1. Mean log10 changes in clindamycin-resistant *C. acnes* counts during 16 weeks of treatment. Data from Cunliffe et al. (2002); visualization created using Python.

3.2 Antibiotic Resistance Across Published Studies

Using data from 23 studies involving 2,046 *C. acnes* isolates, I calculated descriptive resistance percentages for 11 antibiotics. Among the antibiotics commonly used to treat acne, erythromycin (35.33%) and clindamycin (27.98%) showed higher resistance percentages than doxycycline (3.79%) and minocycline (1.57%) (Zhu et al., 2025).

Clindamycin resistance also varied across the 18 studies reporting this antibiotic, ranging from 0% to approximately 56.82%. Differences were observed across countries, although the number of studies and isolates varied by location. These percentages were calculated from published study-level data and are not the pooled estimates reported in the original meta-analysis (Zhu et al., 2025).

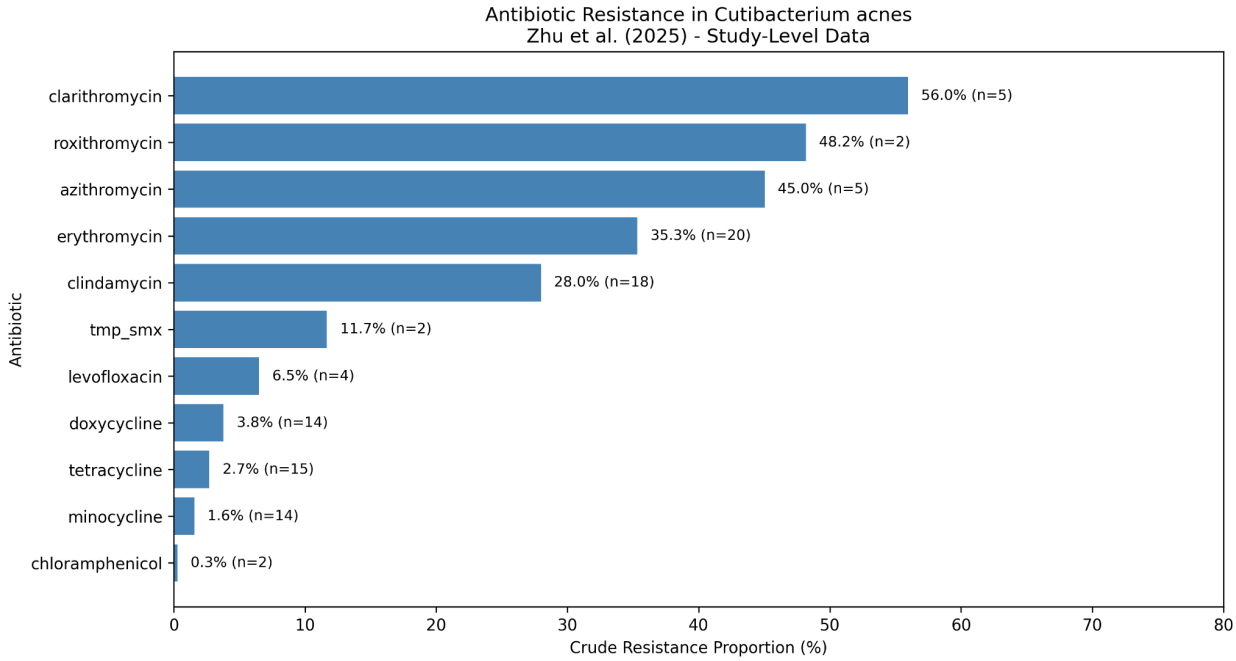


Figure 2. Descriptive resistance percentages for 11 antibiotics. Calculated from published data in Zhu et al. (2025); visualization created using Python.

3.3 Genetic Mutations Associated with Antibiotic Resistance

Using data from Ross et al. (1997), I examined 26 antibiotic-resistant clinical isolates with reported 23S rRNA mutations. The most common mutation was A2058G, found in 13 isolates (50.0%), followed by A2059G in 7 isolates (26.9%) and G2057A in 6 isolates (23.1%).

These mutations affect the 23S rRNA region of the bacterial ribosome, where certain antibiotics bind. Changes in this region are associated with resistance to antibiotics such as clindamycin and erythromycin (Ross et al., 1997).

Of the 26 isolates, 22 were *P. acnes* (now called *C. acnes*), while 4 belonged to related bacterial species. Therefore, these percentages describe the selected resistant isolates in the original study, not the overall *C. acnes* population (Ross et al., 1997).

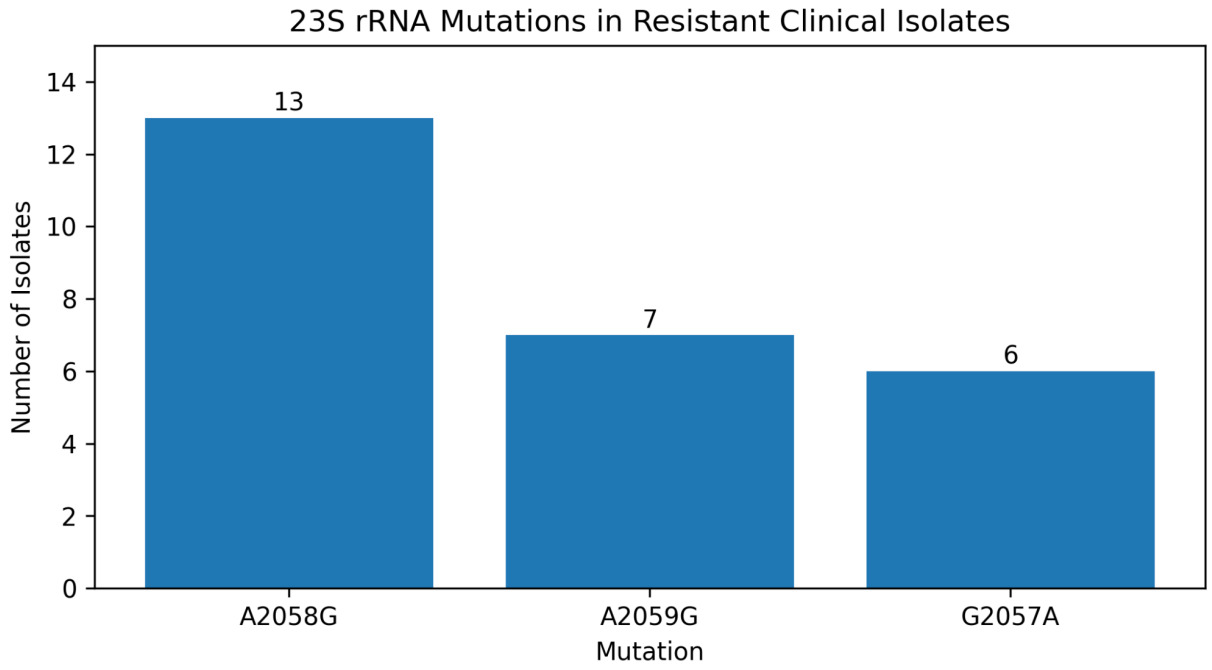


Figure 3. Distribution of 23S rRNA mutations among 26 antibiotic-resistant clinical isolates. Data from Ross et al. (1997); visualization created using Python.

3.4 Topical Combination Treatment and Antibiotic Resistance

Using published data from Ghannoum et al. (2025), I examined changes in *C. acnes* detection and clindamycin resistance among 45 participants who completed 24 weeks of treatment with clindamycin, adapalene, and benzoyl peroxide.

The number of participants with detectable *C. acnes* decreased from 37 (82.2%) at baseline to 20 (44.4%) at Week 24. However, the number of participants with clindamycin-resistant bacteria remained at five. All five participants with resistant bacteria at baseline still had resistant bacteria detected at Week 24, and no newly detected cases of clindamycin resistance were reported (Ghannoum et al., 2025).

Among participants with detectable *C. acnes*, the proportion with resistant bacteria increased from 13.5% (5/37) to 25.0% (5/20). This increase occurred because fewer participants had detectable bacteria overall, not because the number of participants with resistant bacteria increased.

All five participants with resistant bacteria at baseline also showed reductions in acne lesions at Week 12 (Ghannoum et al., 2025).

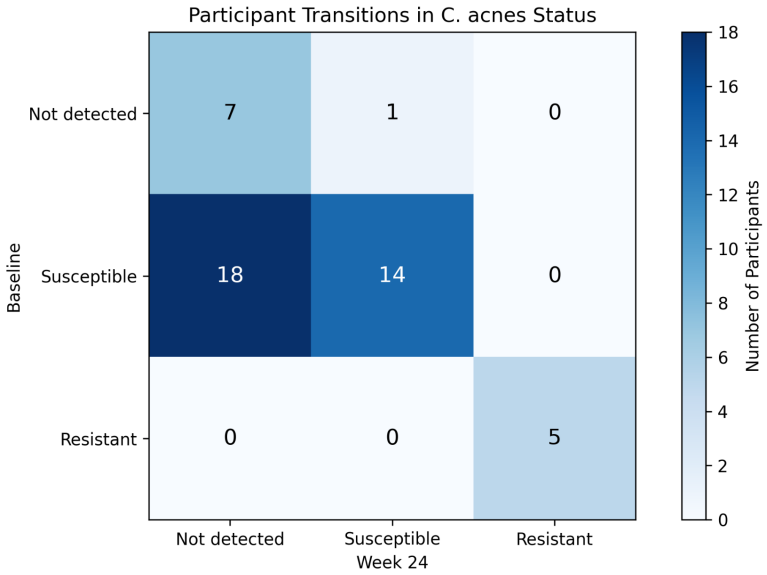


Figure 4. Changes in *C. acnes* detection and clindamycin resistance status from baseline to Week 24 among 45 participants. Data from Ghannoum et al. (2025); visualization created using Python.

4. Discussion

The four analyses suggest that antibiotic resistance in *C. acnes* is associated with several factors, including the antibiotic studied, genetic mutations, and treatment approach.

The resistance patterns calculated from Zhu et al. (2025) suggest that resistance is more frequently reported for some antibiotics than others. Ross et al. (1997) also identified mutations in 23S rRNA associated with resistance. Together, these findings show why understanding both resistance rates and bacterial genetics is important when studying acne treatments.

Treatment choice may also matter. The analysis of Cunliffe et al. (2002) showed that combining clindamycin with benzoyl peroxide was associated with lower resistant bacterial counts and greater acne improvement than clindamycin alone. This supports the potential benefits of combination treatment, although the findings come from a single clinical trial.

The longer-term data from Ghannoum et al. (2025) provide another perspective. Although fewer participants had detectable *C. acnes* after treatment, the five participants with resistant bacteria at baseline remained resistant at Week 24. This suggests that reducing bacterial detection does not necessarily eliminate existing antibiotic resistance.

Overall, these findings suggest that combination treatments may offer advantages over topical antibiotic use alone. However, the available data do not establish that combination treatment prevents resistance in all patients. Further studies comparing different treatment approaches over longer periods are needed.

5. Limitations

This project has several limitations. First, all four analyses used published data rather than original patient-level data. The results are descriptive and cannot establish cause-and-effect relationships.

Second, the studies differed in design, sample size, and methods. The resistance percentages calculated from Zhu et al. (2025) were based on reported study-level data and should not be confused with the original meta-analysis estimates. The genetic analysis from Ross et al. (1997) included selected resistant isolates, so its mutation frequencies do not represent the general *C. acnes* population.

Finally, the treatment findings have limitations. Cunliffe et al. (2002) examined treatment outcomes in one clinical trial, while Ghannoum et al. (2025) reported results from two pooled studies without a control group. Therefore, these findings cannot establish whether combination treatment prevents antibiotic resistance over time.

6. Conclusion

This independent research project examined antibiotic resistance in *Cutibacterium acnes* through four descriptive analyses of published data. The findings showed that resistance levels differed across antibiotics and studies, and that mutations in 23S rRNA were associated with resistance (Ross et al., 1997; Zhu et al., 2025).

The treatment analyses suggested that combining clindamycin with benzoyl peroxide may offer advantages over clindamycin alone, including lower resistant bacterial counts and greater improvement in acne lesions (Cunliffe et al., 2002). However, resistant bacteria remained detectable in some participants during longer-term combination treatment (Ghannoum et al., 2025). Together, these findings highlight the importance of considering antibiotic resistance when choosing acne treatments and the need for further research on effective alternatives.

7. References

Cunliffe, W. J., Holland, K. T., Bojar, R., & Levy, S. F. (2002). A randomized, double-blind comparison of a clindamycin phosphate/benzoyl peroxide gel formulation and a matching clindamycin gel with respect to microbiologic activity and clinical efficacy in the topical treatment of acne vulgaris. *Clinical Therapeutics, 24*(7), 1117–1133. https://doi.org/10.1016/S0149-2918(02)80023-6

Ghannoum, M., Eltokhy, A., Sewake, J., McCormick, T., Bhatia, N., Baldwin, H., Stein Gold, L., Harper, J. C., Zeichner, J. A., Lain, E. T., Callender, V. D., Guenin, E., & Draelos, Z. D. (2025). Assessing *Cutibacterium acnes* susceptibility with clindamycin phosphate 1.2%/adapalene 0.15%/benzoyl peroxide 3.1% gel use: Six-month analysis. *Journal of Drugs in Dermatology, 24*(11), 1106–1111. https://doi.org/10.36849/JDD.9213

Ross, J. I., Eady, E. A., Cove, J. H., Jones, C. E., Ratyal, A. H., Miller, Y. W., Vyakrnam, S., & Cunliffe, W. J. (1997). Clinical resistance to erythromycin and clindamycin in cutaneous propionibacteria isolated from acne patients is associated with mutations in 23S rRNA. *Antimicrobial Agents and Chemotherapy, 41*(5), 1162–1165. https://doi.org/10.1128/AAC.41.5.1162

Zhu, C., Wei, B., Li, Y., & Wang, C. (2025). Antibiotic resistance rates in *Cutibacterium acnes* isolated from patients with acne vulgaris: A systematic review and meta-analysis. *Frontiers in Microbiology, 16*, Article 1565111. https://doi.org/10.3389/fmicb.2025.1565111